

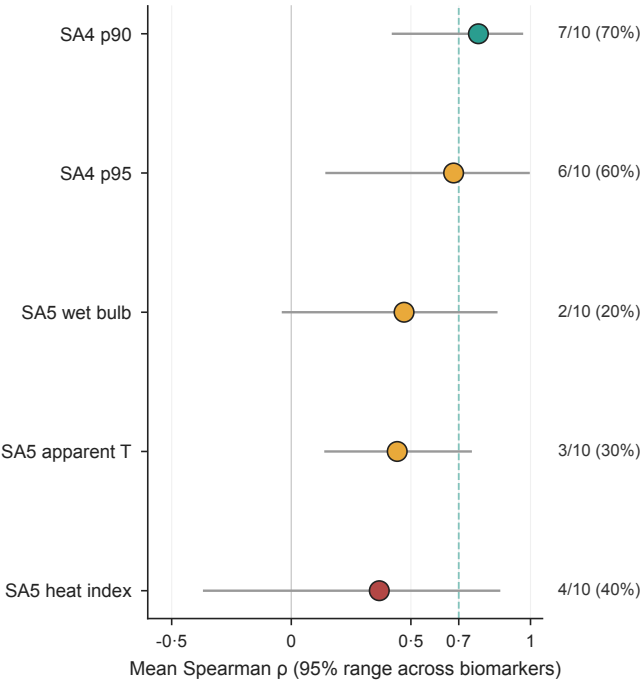
Concordance of each sensitivity specification with the primary Stage 1 model.
Findings are robust to heat-wave threshold (SA4 p95/p90) but sensitive to exposure-metric choice (SA5 humidity-aware indices) and to HIV-status stratification (SA3 panel d).

* $\rho \geq 0.70$ counted as concordant (bold outline in panel a).

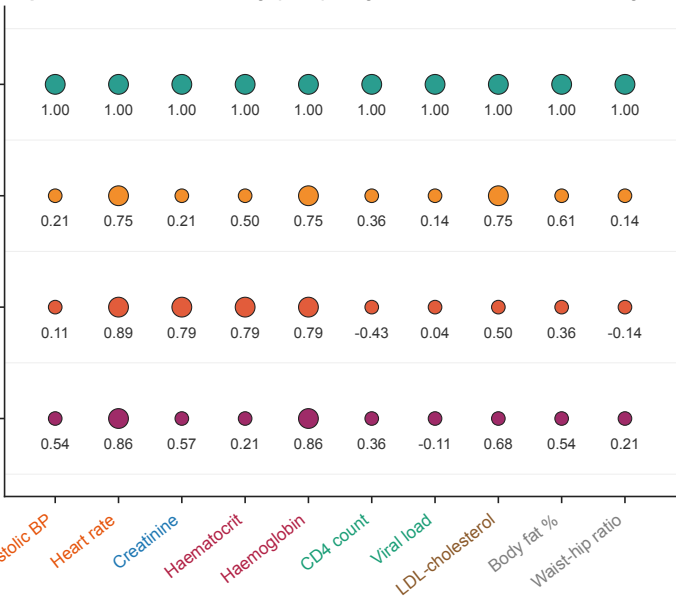
Spearman ρ of SHAP lag profile vs primary specification



Sensitivity ranking



Exposure-metric sensitivity (SA5) - dry-bulb reference vs humidity-aware indices



SA3 | HIV status as effect modifier

7 of 10

retained biomarkers show distinct lag-response profiles by HIV status

Effect modifiers (HIV+ vs HIV- $\rho < 0.70$):

- Body fat %
- Waist-hip ratio
- Creatinine
- Diastolic BP
- Haematocrit
- Haemoglobin
- LDL-cholesterol

Not modified:

- Heart rate

No HIV-negative comparator: CD4 count, Viral load

→ motivates HIV-stratified primary analysis.